

University of Nottingham
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COMP3041 Coursework Part 1

Ethical Investigation of the iBilik Rental Ecosystem

Group A

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1. Introduction and System Selection

For this ethical investigation, our group selected iBilik, a Malaysian accommodation and rental platform. Its website enables users to search for rooms, properties and short-term accommodation, while iBilikPlus provides tenancy-management and rental-payment functions.

iBilik was selected because it is a real, accessible and locally relevant platform that can be investigated directly. Unlike widely studied global platforms, its ethical implications have received comparatively little academic attention. On a more personal level, as final-year students close to graduation, most of us will soon need to find rented rooms, often in a new city and on a limited budget. Platforms like iBilik are likely to be our first stop, so we have a direct interest in whether they treat renters fairly, protect their data and present costs honestly.

iBilik's software design may nevertheless have significant consequences: search filters can affect access to accommodation, verification features can influence trust, data-processing practices can affect privacy, and financial terms can affect users after they commit money to the service.

Our preliminary investigation identified four issues: discrimination and unequal access to housing; safety, trust and platform responsibility; privacy, profiling and personal-data use; and financial fairness, transparency and accountability. Part 1 establishes evidence for these concerns, connects them with relevant literature, and proposes an ethical framework for deeper analysis in Part 2.

2. Preliminary Investigation Method

The investigation combined direct platform testing, policy analysis, technical inspection and publicly available evidence.

Hands-on walkthroughs of iBilik and iBilikPlus examined accommodation filters, listing preferences, registration and verification processes, and financial features. Screenshots were collected to support observations with first-hand evidence.

The group also examined iBilik's Terms of Service, iBilikPlus Terms and Conditions, and privacy documentation (iBilik, n.d.-a, n.d.-b; iBilikPlus, n.d.). These were compared with observed platform behaviour. Public reports and complaints concerning scams, misleading listings and financial loss were used as supporting evidence rather than substitutes for our own investigation.

A focused privacy investigation on 25 October 2026 examined cookies and third-party requests generated when the homepage first loaded. The HTML source of 20 publicly accessible room listings was also analysed without logging in. Combining these methods allowed observed behaviour to be compared with the platform's stated policies.

As this was a preliminary investigation, the sampled listings and observed platform behaviour provide evidence of potential ethical concerns but do not establish how frequently these issues occur across the entire iBilik ecosystem.

3. Preliminary Ethical Issues Identified

3.1 Discrimination and Unequal Access to Housing

The first ethical issue concerns how iBilik's platform design may facilitate unequal access to housing. Its search interface allows accommodation to be filtered by characteristics including race, nationality, relationship status and occupation (Figure 1). These are structured platform options rather than free-text preferences entered only by landlords.

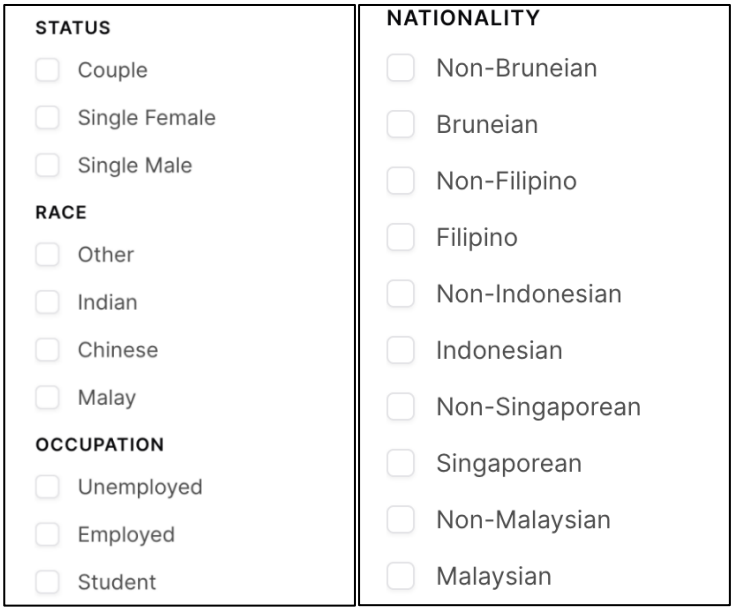


Figure 1: Demographic Search Filters Available on iBilik

Our investigation found substantial differences in listing availability after race filters were applied. In Cheras, 1,834 listings were initially available, compared with 590 for Indian, 897 for Malay, 1,038 for Chinese and 493 for Other. In Bukit Jalil, 986 listings were initially available, compared with 175, 308, 395 and 243 respectively. Moreover, in Petaling Jaya, 6314 listings were available. Under the Indian and Other filters, only 1522 and 1548 listings were available respectively, compared with 3276 for Malay and 3533 for Chinese.

Table 1: Availability of Listings in Cheras, Bukit Jalil, and Petaling Jaya After Race Filtering

	Cheras (1834)	**Bukit Jalil (986)**	**Petaling Jaya (6314)**
Chinese	57%	40%	56%
Malay	49%	31%	51%
Indian	32%	18%	24%
Other	27%	25%	25%

Differential treatment was also observed at listing level. One listing offered zero deposit to Malaysians but required a one-month deposit from non-Malaysians (Figure 2), while another explicitly preferred a "single Chinese female" (Figure 3).



Figure 2: Nationality-Based Difference in Deposit Requirements

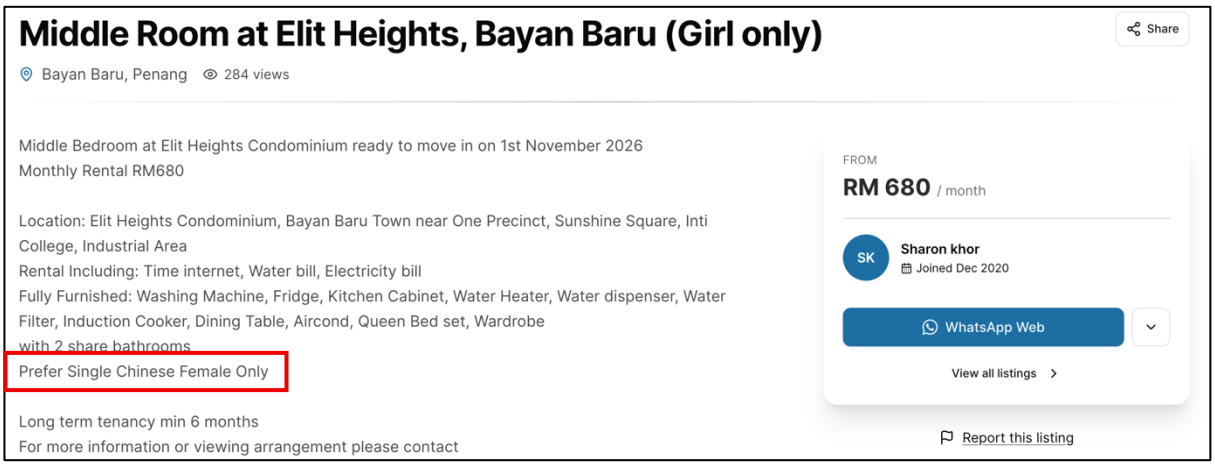


Figure 3: Example of a Listing Specifying a Racial and Gender Preference

The ethical concern is therefore not limited to individual landlord behaviour. iBilik's structured interface makes demographic preferences easier to express and apply. However, different preferences may require different ethical treatment; for example, gender preferences in shared accommodation may involve privacy or safety considerations, while race-based restrictions are harder to justify on the same basis. This distinction will be examined further in Part 2.

3.2 Safety, Trust and Platform Responsibility

The second issue concerns the level of responsibility iBilik accepts for safety and trust on its platform. Public user reports describe rental scams involving fraudulent listings, including one case where a user reportedly lost RM700. The listing involved was marked as having no verified details and advised users to contact the owner at their own risk (Figure 4) (Anonymous participant, 2024).

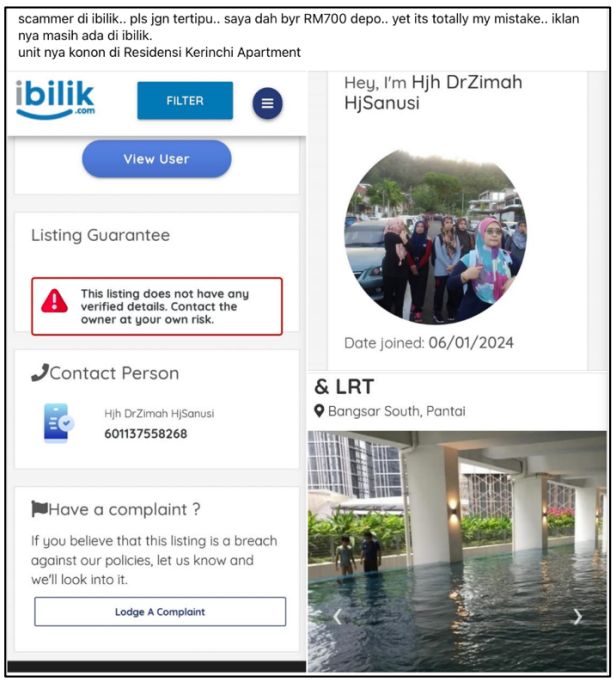


Figure 4: Reported RM700 Rental Scam Involving an Unverified iBilik Listing

This is significant when compared with iBilik's Terms of Service, which state that the platform does not verify user statements or inspect accommodation and instead expects users to take their own precautions (iBilik, n.d.-b; Figure 5).

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Figure 5: iBilik Disclaimer Regarding User Verification and Accommodation Inspection

At the same time, some advertisers display a “KYC Verified” badge. Our investigation found that this appears to involve identity documents such as an IC or passport, but identity verification does not necessarily confirm property ownership or authority to rent (Figure 6).

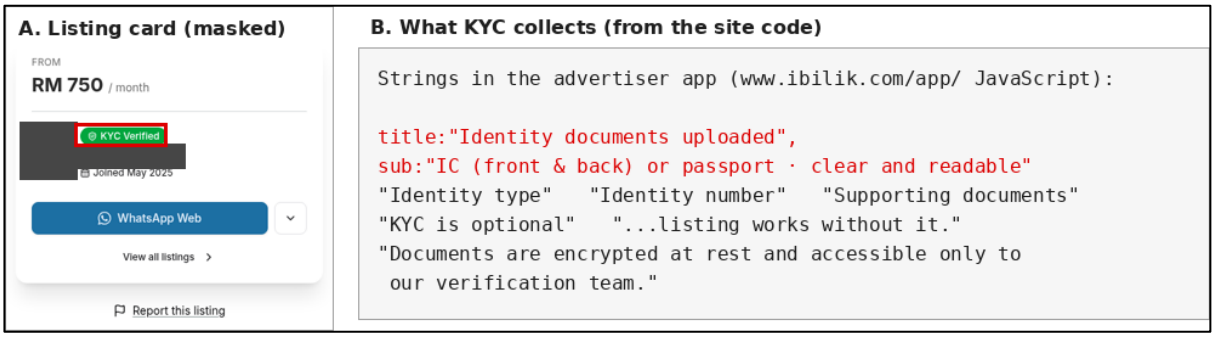


Figure 6: iBilik “KYC Verified” Badge and Advertiser Identity-Document Requirements

This creates a trust issue because verification indicators may increase user confidence even though the platform limits responsibility for the legitimacy of listings. The ethical question is whether the safeguards provided are proportionate to the trust these indicators may encourage.

3.3 Privacy, Profiling and Personal-Data Use

The third issue concerns the amount of personal information processed through iBilik and iBilikPlus, together with the level of transparency and control provided to users.

The iBilikPlus Terms and Conditions permit the processing of extensive personal information, including demographic, identity, contact, financial and employment information (iBilikPlus, n.d.; Figure 7 & 8).

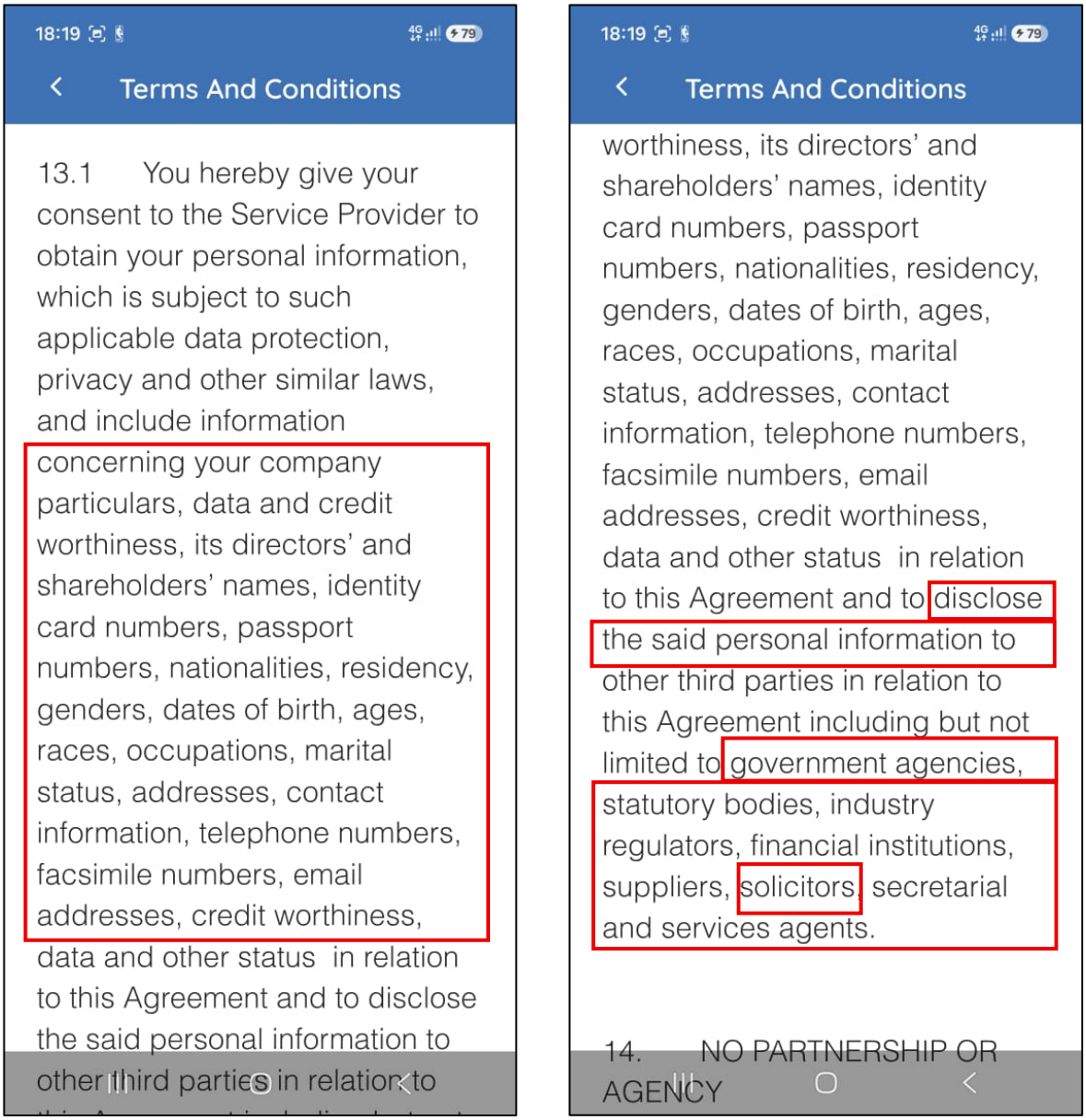
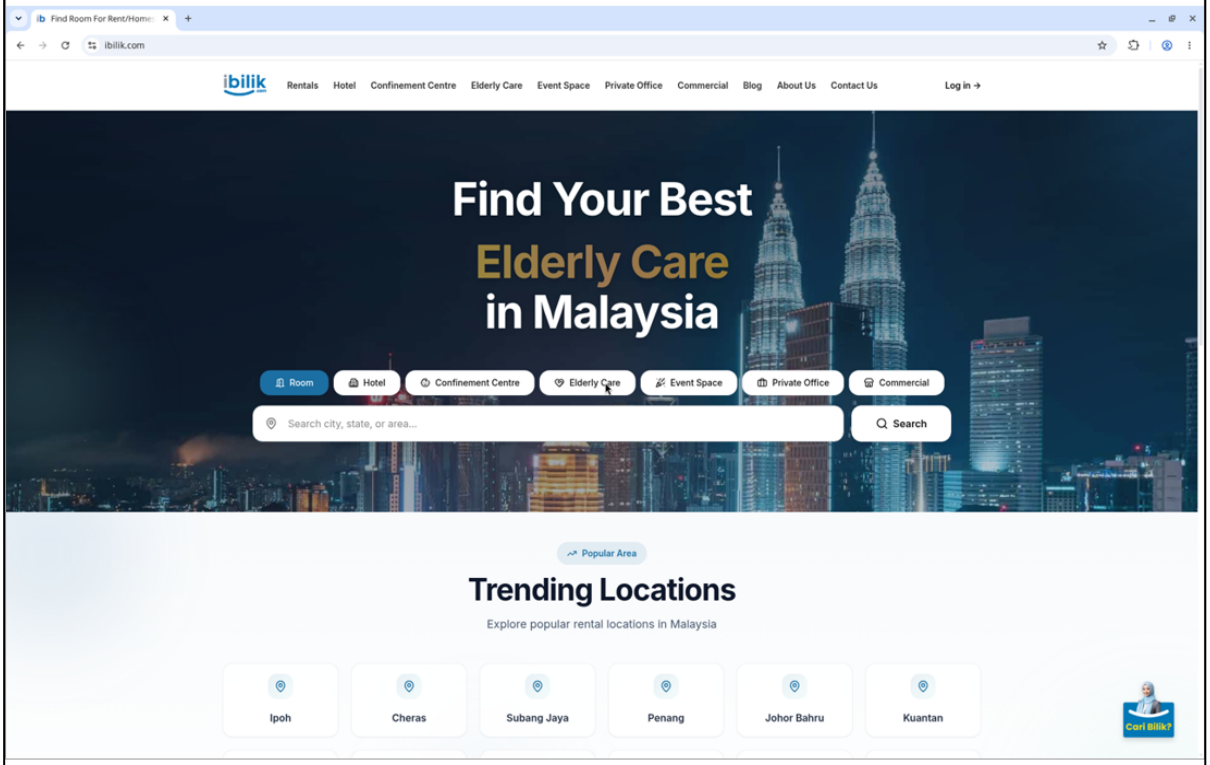


Figure 7 and Figure 8: Categories of Personal Information Processed by iBilikPlus

Some of this information may be necessary for tenancy management or identity verification, but the breadth of collection raises questions about proportionality and data minimisation.

Further concerns were identified during technical testing. Although iBilik's Privacy Policy states that the platform honours Do Not Track signals (iBilik, n.d.-a), Google Analytics and Google Ads-related cookies were recorded on initial page load without a visible cookie banner or consent prompt, and the site's code contains no Do Not Track check (Figure 9). Further investigation showed up to 71 Google Analytics and Google Ads-related cookies tracked in the booking page (Figure 10).

A. www.ibilik.com on first load (25 Oct 2026): no cookie banner or consent prompt



B. Cookies and requests recorded on first load (our browser log, re-typed as text)

```
Recorded in the browser on first load, before any click or consent:

Cookies set by ibilik.com:   _ga   _ga_218372RVB0   _gcl_au
   (_ga = Google Analytics;  _gcl_au = Google Ads conversion linker)

Third-party hosts contacted: www.googletagmanager.com, analytics.google.com,
                             www.google.com, static.cloudflareinsights.com
```

Figure 9: iBilik Homepage on First Load: No Consent Prompt and Google Cookies Set

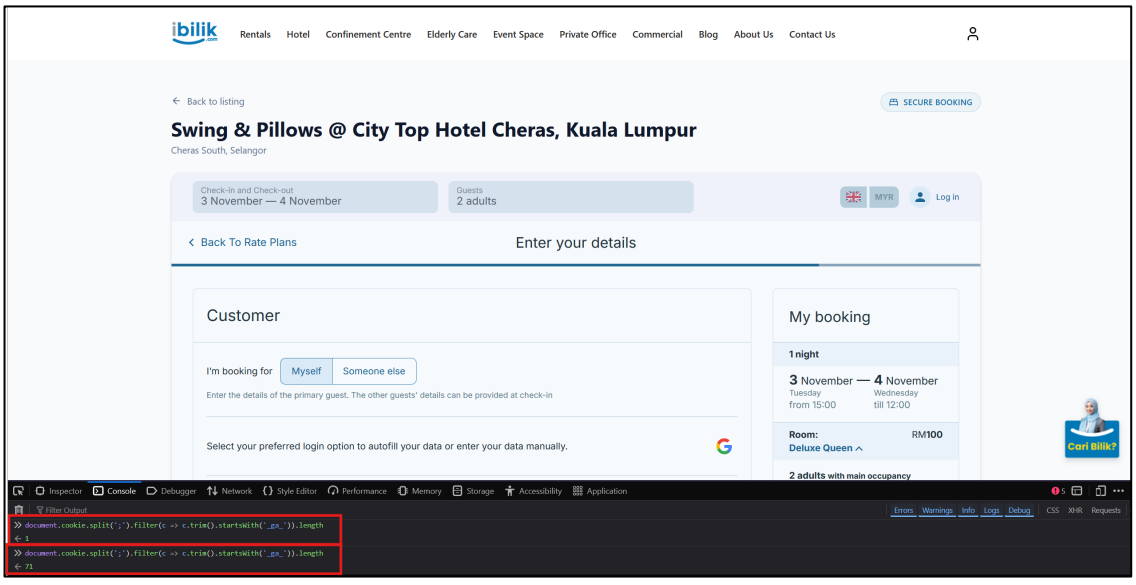


Figure 10: Google Analytics Cookies Set on the Homepage (1) and Booking Page (71), Measured in the Browser Console

The registration process also raises questions about informed consent. The sign-up form contained no explicit checkbox requiring agreement to the Privacy Policy or Terms of Service. Instead, the Privacy Policy states that users are "taken to have agreed" to data processing through use of the platform (iBilik, n.d.-a; Figure 11).

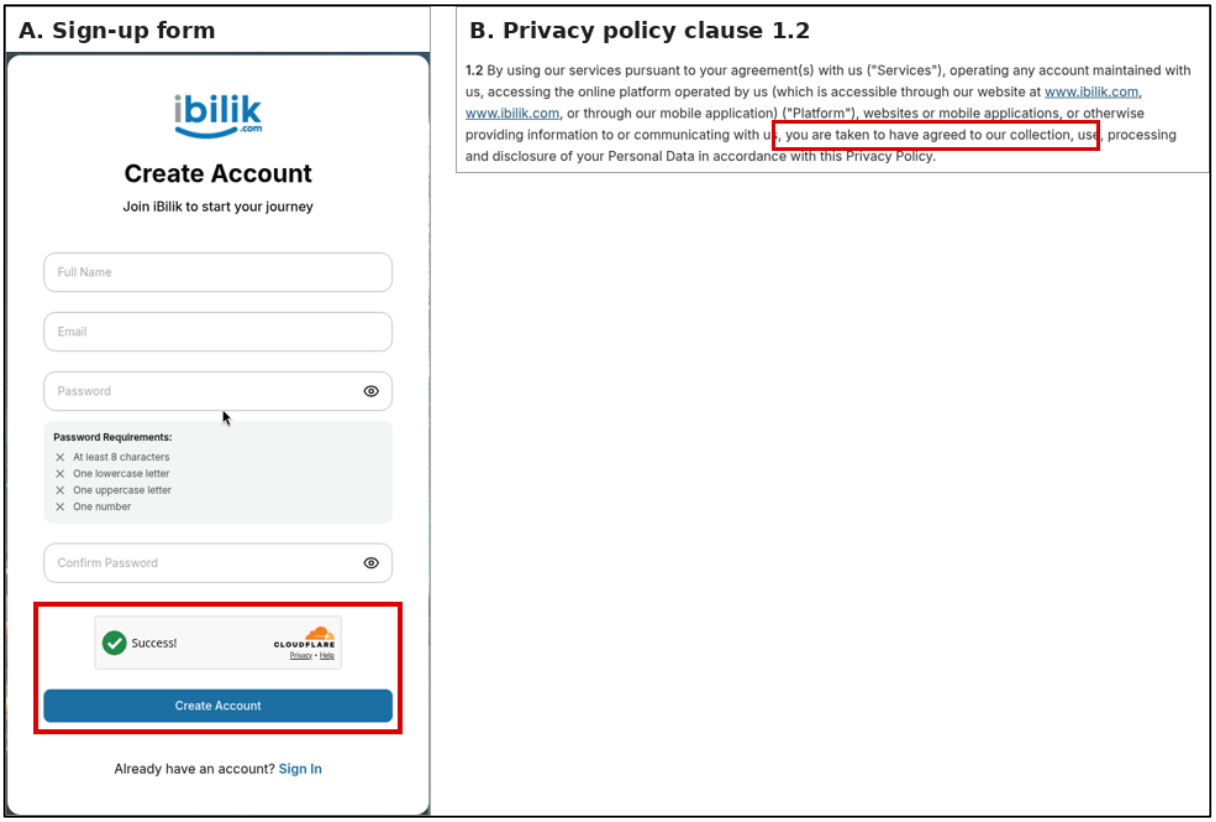


Figure 11: iBilik Registration Form and Implied Privacy-Policy Consent

Twenty public room listings were also checked without logging in. Although each listing displays only a short display name, the page's underlying code also contains the advertiser's phone number and, in most cases, a different full name (Table 2).

Table 2: Personal Data Found in the Page Code of 20 Sampled Listings

Listing	**Phone Number in Page Code**	**Full Name Differs from Displayed Name**
1	Yes (60102XXXX47)	Yes
2	Yes (60163XXXX14)	Yes
5	Yes (60112XXXX07)	No
6	Yes (60116XXXX41)	Yes
All 20	**20 of 20**	**16 of 20**

Together, these findings raise concerns about transparency, meaningful consent and data minimisation.

3.4 Financial Fairness, Transparency and Accountability

The final issue concerns whether iBilikPlus provides users with sufficient clarity and control

over financial conditions.

The iBilikPlus Terms and Conditions allow the service provider to revise subscription fees, credit rates and server maintenance fees, with users given seven days to decide whether to terminate or continue the agreement (iBilikPlus, n.d.; Figure 12). They also state that payments are non-refundable and non-transferable (iBilikPlus, n.d.; Figure 13).

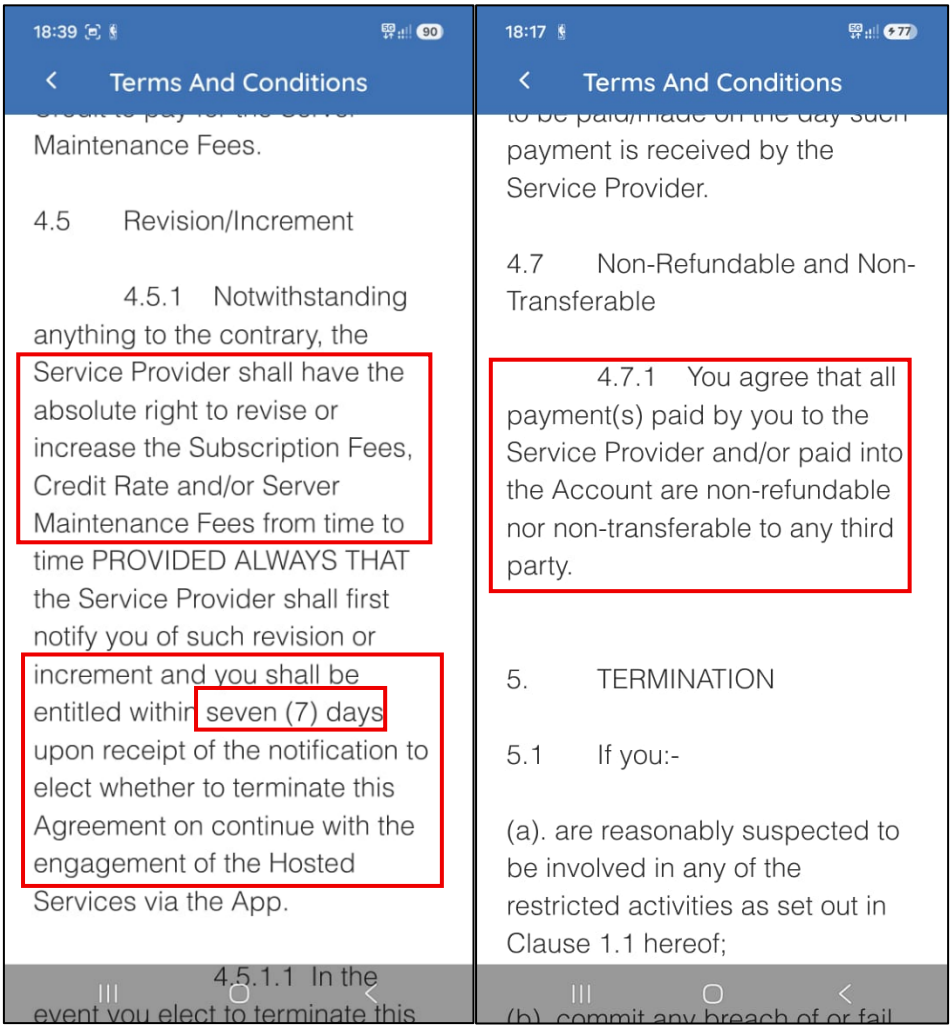
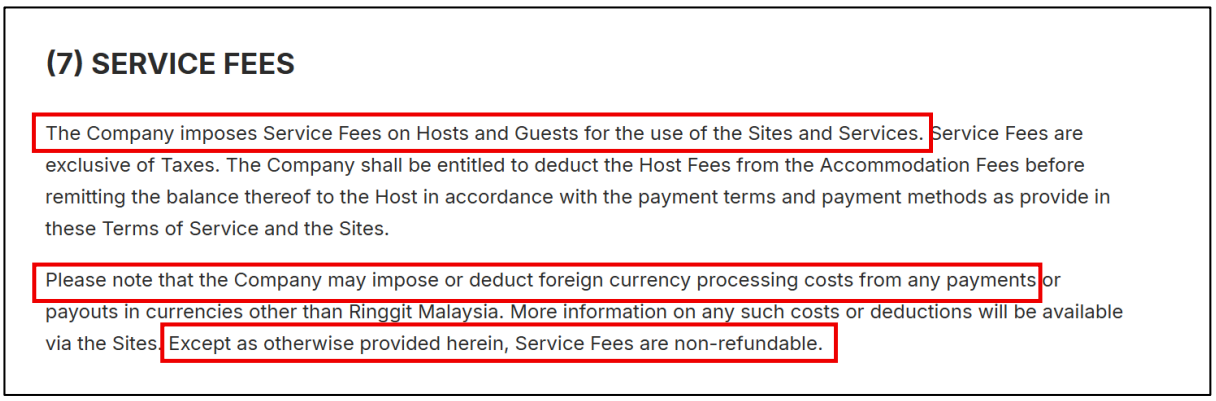


Figure 12 and Figure 13: iBilikPlus Clauses Permitting Revision or Increase of Service Charges, and Stating That Payments Are Non-Refundable and Non-Transferable

Similar restrictions appear in iBilik's Terms of Service, where service fees may apply to both hosts and guests, foreign-currency processing costs may be imposed, and service fees are generally non-refundable (iBilik, n.d.-b; Figure 14).



(7) SERVICE FEES

The Company imposes Service Fees on Hosts and Guests for the use of the Sites and Services. Service Fees are exclusive of Taxes. The Company shall be entitled to deduct the Host Fees from the Accommodation Fees before remitting the balance thereof to the Host in accordance with the payment terms and payment methods as provide in these Terms of Service and the Sites.

Please note that the Company may impose or deduct foreign currency processing costs from any payments or payouts in currencies other than Ringgit Malaysia. More information on any such costs or deductions will be available via the Sites. Except as otherwise provided herein, Service Fees are non-refundable.

Figure 14: iBilik Website Terms Concerning Service Fees and Non-Refundable Charges

These conditions are not automatically unethical, since platforms may reasonably charge fees and revise prices. The ethical concern is whether users receive sufficient notice, understanding and practical control over important financial consequences before and after committing to the service.

4. Literature Review

4.1 Platform Design and Discrimination

Research on rental and sharing platforms shows that discrimination can arise not only from individual users but also from platform design. Meers (2024b) found that built-in classifications and preference inputs on a rental platform facilitated exclusion, while Meers (2024a) showed that "structuring" and "sorting" interface functions can either increase or reduce discriminatory outcomes. A broader systematic review similarly identified user-profile features as recurring mechanisms through which discrimination can emerge across sharing-economy platforms (Tushev et al., 2022). Together, these studies suggest that iBilik's demographic filters should be examined as design choices that may shape access to housing rather than as neutral tools.

4.2 Safety, Trust and Platform Responsibility

Trust is particularly important where platforms mediate transactions between strangers. Soleimani's (2022) review identifies privacy, security, reputation and website characteristics as important influences on trust and perceived risk, while distinguishing trust in individual users from trust in the platform itself. This distinction is relevant to iBilik because a "KYC Verified" badge may increase confidence in an advertiser even when the platform limits responsibility for the accuracy or legitimacy of listings.

4.3 Privacy, Consent and User Control

Privacy research emphasises that publishing a policy alone does not guarantee meaningful transparency. Javed and Sajid (2024) focus on whether privacy policies are accessible, readable, complete and consistent with actual data practices, while Burkhardt et al. (2023) extend this concern to the quality of consent, arguing that users should receive relevant information without manipulation or coercion. Biega et al. (2020) add a further perspective through data minimisation, requiring personal information to be limited to what is necessary for its stated

purpose. Together, these perspectives are relevant to iBilik's broad data collection, implied consent and observed third-party tracking.

4.4 Financial Fairness and Transparency

Research shows that the presentation of financial information can materially influence consumer decisions. Blake et al. (2021) demonstrate specifically that obscured mandatory fees make price comparison more difficult and can increase consumer spending, while Luguri and Strahilevitz (2021) provide a broader explanation by showing how interface techniques such as hidden information and obstruction can influence users toward decisions they may not otherwise make. Together, these findings suggest that financial fairness depends not only on whether fees and restrictions are disclosed, but also on whether their consequences are sufficiently visible and understandable before users commit.

4.5 Literature Synthesis

Across the four themes, the literature indicates that online platforms are not neutral intermediaries: interface design, verification signals, privacy practices and information presentation can materially influence user outcomes. This provides a basis for assessing whether iBilik's observed features are ethically justified in Part 2.

5. Proposed Ethical Framework

The primary professional framework proposed for Part 2 is the *Software Engineering Code of Ethics and Professional Practice* developed by the ACM and IEEE Computer Society (ACM/IEEE-CS Joint Task Force, 1999). Its principles are relevant because the issues identified in iBilik concern consequences arising from software design as well as business policy.

For discrimination and unequal access, Clauses 1.02 and 1.07 are relevant because they concern the public good and factors that may restrict access to software's benefits. Clause 3.03 further requires consideration of foreseeable ethical, economic, cultural and legal issues. These clauses provide a basis for assessing iBilik's structured demographic filters.

For safety and platform responsibility, Clauses 1.03 and 3.10 are relevant to questions of software safety, adequate testing and foreseeable risks. They can therefore be applied to verification indicators, misleading listings and the safeguards provided to users.

For privacy, Clauses 3.12 and 3.13 concern respect for privacy, ethical and authorised use of data. These clauses are relevant to iBilik's extensive data collection, third-party tracking, public exposure of information.

For financial fairness, Clauses 1.06 and 6.07 concern fairness, accurate representation and avoidance of misleading claims. They provide a basis for evaluating how refund restrictions, fees and changes in charges are communicated.

However, the Code may not fully resolve every dilemma. This is particularly relevant to demographic preferences. While the Code supports fairness and equal access, it may not determine whether all preferences should be treated identically. For example, a gender

preference in shared accommodation may relate to privacy or safety, whereas race-based exclusion may lack an equivalent justification.

Two theories will supplement the Code where it falls short. Deontology suits the consent and race-filter issues, because implied consent and race-based exclusion fail to treat users with equal respect, whatever their outcomes. Utilitarianism suits issues involving genuine trade-offs, such as gender preferences, identity verification and fee terms, where benefits to some must be weighed against harms to others. Where the two theories disagree, Part 2 will assess which reasoning is more defensible.

6. Conclusion

The preliminary investigation establishes iBilik as a suitable case for deeper ethical analysis, with first-hand evidence concerning discrimination, platform trust, privacy and financial fairness. Relevant literature indicates that these concerns may arise from both user behaviour and platform design. Part 2 will build on these findings by applying relevant SE Code provisions, supplemented by deontology and utilitarianism where the professional framework alone is insufficient.

References

ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices. (1999). *Software engineering code of ethics and professional practice* (Version 5.2). Association for Computing Machinery. https://ethics.acm.org/code-of-ethics/software-engineering-code/

Anonymous participant. (2024, February 2). scammer di ibilik.. pls jgn tertipu.. saya dah byr RM700 depo.. yet its totally my mistake.. iklan nya masih ada… [Status update]. Kerinchi Residensi Community. Facebook. https://www.facebook.com/groups/1749380641953557/posts/3419550484936556/

Biega, A. J., Potash, P., Daumé III, H., Diaz, F., & Finck, M. (2020). Operationalizing the legal principle of data minimization for personalization. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 399–408). Association for Computing Machinery. https://doi.org/10.1145/3397271.3401034

Blake, T., Moshary, S., Sweeney, K., & Tadelis, S. (2021). Price salience and product choice. *Marketing Science, 40*(4), 619–636. https://doi.org/10.1287/mksc.2020.1261

Burkhardt, G., Boy, F., Doneddu, D., & Hajli, N. (2023). Privacy behaviour: A model for online informed consent. *Journal of Business Ethics, 186*(1), 237–255. https://doi.org/10.1007/s10551-022-05202-1

iBilik. (n.d.-a). *Privacy policy*. Retrieved November 3, 2026, from https://www.ibilik.com/privacy

iBilik. (n.d.-b). *Terms of service*. Retrieved November 3, 2026, from https://www.ibilik.com/term

iBilikPlus. (n.d.). *Terms of use & privacy policy*. Retrieved November 3, 2026, from https://rms.ibilik.com/privacy-policy

Javed, Y., & Sajid, A. (2024). A systematic review of privacy policy literature. *ACM Computing Surveys, 57*(2), Article 45. https://doi.org/10.1145/3698393

Luguri, J., & Strahilevitz, L. J. (2021). Shining a light on dark patterns. *Journal of Legal Analysis, 13*(1), 43–109. https://doi.org/10.1093/jla/laaa006

Meers, J. (2024a). Discrimination at the interface: The Equality Act 2010 and platform interface design. *The Modern Law Review, 87*(3), 640–669. https://doi.org/10.1111/1468-2230.12855

Meers, J. (2024b). "Professionals only please": Discrimination against housing benefit recipients on online rental platforms. *Housing Studies, 39*(1), 29–51. https://doi.org/10.1080/02673037.2021.2014416

Soleimani, M. (2022). Buyers' trust and mistrust in e-commerce platforms: A synthesizing literature review. *Information Systems and e-Business Management, 20*(1), 57–78. https://doi.org/10.1007/s10257-021-00545-0

Tushev, M., Ebrahimi, F., & Mahmoud, A. (2022). A systematic literature review of anti-discrimination design strategies in the digital sharing economy. *IEEE Transactions on Software Engineering, 48*(12), 5148–5157. https://doi.org/10.1109/TSE.2021.3139961